

GREYSON MEARES

Morgantown, WV | (304) 282-8498 | gcm00010@mix.wvu.edu | <https://greyson-m.github.io>

EDUCATION

West Virginia University — Morgantown, WV
M.S. in Computer Science

Aug 2026 - Present

West Virginia University — Morgantown, WV
B.S. in Computer Science | Minor in Applied Mathematics

Aug 2022 - May 2026

TECHNICAL SKILLS

Programming: C++, C, Python, MATLAB, Java, JavaScript

Systems/Tools: OpenGL, Git, Linux

Math/Computing: Numerical methods, Algorithms, Spectral methods

Web/App: React, Node.js, Flask, MongoDB

PUBLICATION

Undergraduate Research, Department of Mathematics — West Virginia University May 2025 – Feb. 2026

- **Publication:** Greyson Meares, Sage Meiling, and Charis Tsikkou. "Dynamic Response of a Finite Circular Plate on an Elastic Half-Space Using the Truncated Lamb Kernel." *International Journal of Engineering Science*, <https://www.sciencedirect.com/science/article/pii/S0020722526000844>
- Developed and implemented numerical methods (spectral/Galerkin techniques, Hankel/Fourier transforms, careful quadrature) to solve and study a PDE model
- Validated against limiting cases and benchmarked numerical stability/conditioning
- Tools: Python (NumPy, SciPy, Matplotlib), reproducible workflows, version control

PROJECTS - <https://greyson-m.github.io>

- Dzhaniybekov Effect Simulation: Numerical rigid-body dynamics with a custom **3D renderer using OpenGL**
- Computational Fluid Dynamics: Developed code for solving 2D Euler flows
- Nonlinear Conservation-Law Solvers: Implemented and compared schemes; stability/accuracy checks

WORK EXPERIENCE

West Virginia University — Graduate Research Assistant

Aug 2026 – Present

- Conduct research in theoretical computer science
- Prove properties of algorithms in combinatorial optimization and analyze complexity
- Work alongside other students, researchers, and faculty in a collaborative research setting

Lowe's — Electrical/Plumbing Associate — Morgantown, WV

Jul 2023 – Aug 2026

- Apply technical reasoning to customer problems in electrical and plumbing; ensured safety/compliance
- Use enterprise systems to monitor inventory and place orders; maintained organized, searchable stock
- Built product knowledge across categories; communicated clearly with DIY and professional customers

RELEVANT COURSEWORK

Applications of Neural Networks (Grad); Advanced Analysis of Algorithms (Grad); Probability & Statistics; Dynamical Systems; Partial Differential Equations; Real Analysis; Discrete Mathematics 2 (Game Theory); Complex Variables; Linear Algebra; Artificial Intelligence; Microprocessor Systems; Algorithms; Compiler Construction; Data Structures; Operating Systems Structure; Quantum Computing;